

Tony Lam

+1 (540) 815-9070 | lam.tony540@gmail.com | linkedin.com/in/lamtonyys | github.com/tonylam0

EDUCATION

University of Virginia

Bachelor of Arts in Computer Science

Charlottesville, VA

August 2025 – May 2029

- **Relevant Coursework:** Data Structures & Algorithms, Software Development Essentials, Discrete Mathematics

PROJECTS

Evergreen | *React, Django, REST APIs, SQLite*

November 2025 – Present

- Developing a **full-stack** web platform for sharing and discussing video essays, utilizing a decoupled architecture.
- Architected a **Django** backend to serve **RESTful APIs**, handling user authentication and content management.
- Building a responsive frontend with **React** and **Vite**, implementing component-based state management for a seamless user experience.
- Designed a relational database schema to efficiently store user profiles, media content, and comment threads.

Handwritten Digits Recognition | *Python, NumPy*

July 2025 – August 2025

- Developed a **neural network** from scratch to classify handwritten digits using Python and NumPy.
- Implemented core machine learning algorithms including **feedforward propagation**, **backpropagation**, and **gradient descent** without external ML frameworks.
- Trained the model on the **MNIST dataset**, achieving over **95% accuracy** in digit recognition.

Planetary Simulation | *Python, Pygame*

December 2024 – December 2024

- Created a **physics-based simulation** of the inner solar system, modeling orbital mechanics and gravitational attraction.
- Implemented **Object-Oriented Programming (OOP)** principles to design modular Planet and Moon classes with unique physical properties.
- Developed an interactive user experience with dynamic zooming, camera tracking, and orbital path visualization.

Conway's Game of Life | *Python, Pygame, Matplotlib*

July 2025 – July 2025

- Built a **cellular automaton** simulation with interactive controls for pausing, resetting, and speed adjustment.
- Engineered **data visualization** using Matplotlib to track population stability across generations.

EXPERIENCE

Stocker/Backroom Associate

Marshalls

July 2023 – August 2025

Roanoke, VA

- Managed backroom inventory logistics for a high-volume retail location, ensuring accurate stock levels and rapid floor replenishment.
- Collaborated within a **10+ person team** to execute store reorganizations and shipment processing under tight deadlines.
- Resolved customer inquiries and issues effectively, maintaining high service standards during peak hours.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, CSS, SQL

Frameworks/Libraries: React, Django, Flask, Pygame, NumPy, Matplotlib

Developer Tools: Git, GitHub, VS Code, Neovim, Unix Command Line, Bash/Zsh, Vite